Week 11: Bootstrap and Resampling

SDS8102

Week 11

Overview. Week 9 obtained reference distributions from Gaussian limits. Week 10 used those distributions to calibrate discrepancies between data and a model. We now ask how to estimate a reference distribution when its shape or its nuisance parameters are unknown. The central idea is to reproduce the sampling error, including its centering, scale, and data-generating restrictions.

We first derive basic large-sample approximations. We then replace an unknown population by its empirical distribution and prove bootstrap validity for sample means and smooth functions of means. Confidence intervals follow by inverting the estimated error distribution. Subsampling extends this approach to statistics with a known, possibly non-Gaussian limit. Permutation and randomization tests use a different source of validity: a symmetry of the null distribution or a known treatment-assignment design. Finally, Edgeworth expansions explain when the bootstrap estimates the first correction to a Gaussian approximation. *A resampling method works when its conditional reference distribution reproduces the relevant sampling law.*

Prerequisites and reading. The law of large numbers (LLN), central limit theorem (CLT), Taylor expansion, Slutsky's theorem, empirical cdfs, and basic hypothesis testing. Efron [1] introduces the bootstrap; van der Vaart [2], Chapter 23, develops its asymptotic theory. Hall [3] treats higher-order accuracy. Politis and Romano [4] develop subsampling; Chung and Romano [5] distinguish exact permutation validity from asymptotic validity for weaker nulls.

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1 The sampling distribution is the target

1.1 Data, parameters, and roots

Let $\mathcal{D}_n = (X_1, \ldots, X_n)$ consist of independent and identically distributed (iid) observations from a probability distribution P. Write $Pf = \mathbb{E}_P f(X)$ and $P_n f = n^{-1} \sum_i f(X_i)$. The empirical probability distribution P_n puts mass $1/n$ at each observation, counting repeated observations with their multiplicities. Its cdf is $F_n(x) = n^{-1} \sum_i \mathbf{1}\{X_i \le x\}$.

The scalar parameter is $\theta = \theta(P)$ and its estimator is $\widehat{\theta}_n$. *A* root *is a function of the data and the parameter whose sampling distribution we want to approximate.* For example,

$$R_n = \sqrt{n}(\widehat{\theta}_n - \theta), \qquad T_n = \frac{\sqrt{n}(\widehat{\theta}_n - \theta)}{\widehat{\sigma}_n}.$$

The quantity $\widehat{\sigma}_n$ estimates the standard deviation of the limiting law of $\sqrt{n}(\widehat{\theta}_n - \theta)$. The estimated standard error of $\widehat{\theta}_n$ is therefore $\widehat{\sigma}_n/\sqrt{n}$. A *pivotal* root has a distribution independent of unknown population features; an *asymptotically pivotal* root has a limiting distribution with that property.

Write $G_n(t) = \mathbb{P}_P(R_n \le t)$. This probability refers to repeated original samples from P. The data are random under $\mathbb{P}_P$. *In contrast, a resampling probability conditions on the realized data.* Keeping these two probabilities separate is essential.

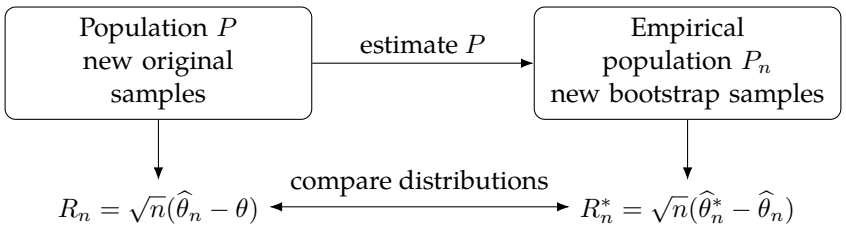
Example 1.1 (A running example: a right-skewed mean)**.** Suppose X has an exponential distribution with mean $\mu > 0$. Then $\mathrm{Var}(X) = \mu^2$ and its standardized third central moment, or *skewness*, is $\gamma = 2$. Our nonparametric analysis does not assume this model when resampling; the model supplies an exact sampling law for illustrating the approximation. For $S_n^2 = n^{-1} \sum_i (X_i - \bar{X})^2$,

$$T_n = \frac{\sqrt{n}(\bar{X} - \mu)}{S_n} \rightsquigarrow N(0, 1).$$

The limit is symmetric, although the finite-sample distribution need not be. This gap motivates the last part of the lecture.

1.2 Three ways to construct a reference distribution

A large-sample approximation replaces G_n by a limiting cdf G. A bootstrap approximation replaces the population by an estimated population and recomputes the root. A permutation or randomization test instead generates a null reference from transformations allowed by a symmetry or by the experimental design (Sections 6.1 and 6.3). *Resampling* methods often cover these procedures, but their validity arguments differ.



Different original samples give different bootstrap distributions. Within one bootstrap analysis, the original sample is held fixed.

2 Basic large-sample approximations

2.1 Linearization, the delta method, and standard errors

Definition 2.1 (Asymptotic linearity)**.** An estimator is asymptotically linear at P if, for a measurable function φ_P,

$$\sqrt{n}(\widehat{\theta}_n - \theta) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \varphi_P(X_i) + o_P(1), \qquad P\varphi_P = 0, \quad 0 < P\varphi_P^2 = \sigma_P^2 < \infty.$$

The function φ_P is an *influence function*: it describes the first-order effect of one observation on the estimator. The notation $o_P(1)$ means convergence to zero in P-probability, and $O_P(1)$ means boundedness in probability.

The CLT and Slutsky give $\sqrt{n}(\widehat{\theta}_n - \theta) \rightsquigarrow N(0, \sigma_P^2)$. If $\widehat{\sigma}_n \xrightarrow{\mathbb{P}} \sigma_P$, then $T_n \rightsquigarrow N(0,1)$. No convergence rate of approximation follows from this limit alone.

A useful class is $\widehat{\theta}_n = h(\bar{Z}_n)$, where $Z_i = g(X_i) \in \mathbb{R}^d$, $\eta = Pg$, and $\theta = h(\eta)$. We regard the dimension d as fixed.

Theorem 2.2 (Smooth functions of means)**.** *Let $X_1, \ldots, X_n$ be iid from P, and set $Z_i = g(X_i) \in \mathbb{R}^d$ for fixed d. Assume $P\|Z\|^2 < \infty$ and $h : \mathbb{R}^d \to \mathbb{R}$ is continuously differentiable in a neighborhood of η. Write $\Sigma = \mathrm{Cov}_P(Z)$, $a = \nabla h(\eta)$, and assume $\sigma^2 = a^{\mathrm{T}}\Sigma a > 0$. Then*

$$\sqrt{n}\{h(\bar{Z}_n) - h(\eta)\} \rightsquigarrow N(0, \sigma^2).$$

With $\widehat{\Sigma}_n = n^{-1}\sum_i (Z_i - \bar{Z}_n)(Z_i - \bar{Z}_n)^{\mathrm{T}}$, then $\widehat{\sigma}_n^2 = \nabla h(\bar{Z}_n)^{\mathrm{T}} \widehat{\Sigma}_n \nabla h(\bar{Z}_n) \xrightarrow{\mathbb{P}} \sigma^2$.

The proof is given in Appendix B.

Remark 2.3 (Intuition). *Near the true mean, a smooth statistic behaves like a weighted average.* Finite second moments give the CLT, and a nonzero limiting variance permits studentization. A zero gradient can change both the rate and the limiting law.

Example 2.4 (A ratio)**.** For $Z_i = (U_i, V_i)^{\mathrm{T}}$, let $\eta = (u, v)^{\mathrm{T}}$ with $v \neq 0$. Taking $h(u, v) = u/v$ gives

$$\varphi_P(U, V) = \frac{U - u}{v} - \frac{u(V - v)}{v^2}.$$

Finite second moments suffice for the first-order result. The requirement $v \neq 0$ ensures local smoothness. A denominator near zero can still make a finite-sample approximation poor.

2.2 Normal intervals and approximation error: A motivation for the bootstrap

Let Φ and ϕ be the standard normal cdf and density, and let $z_p = \Phi^{-1}(p)$ for $0 < p < 1$. The usual two-sided interval is

$$\left[\widehat{\theta}_n - z_{1-\alpha/2}\frac{\widehat{\sigma}_n}{\sqrt{n}}, \quad \widehat{\theta}_n + z_{1-\alpha/2}\frac{\widehat{\sigma}_n}{\sqrt{n}}\right].$$

At each fixed P satisfying the stated conditions, its coverage tends to $1 - \alpha$. This is *pointwise* validity; it does not give a uniform guarantee over distributions approaching a boundary. The follow-

ing Berry–Esseen inequality quantifies the rate of convergence in the CLT. It is a first-order result; higher-order expansions are discussed in Section 7.

Theorem 2.5 (Berry–Esseen bound for an iid mean; this version due to [6]). *Let $X_1, \dots, X_n$ be iid real-valued observations with mean μ, variance $0 < \sigma^2 < \infty$, and $\rho = \mathbb{E}_P|X - \mu|^3 < \infty$. There is a universal constant $C > 0$, independent of P and n, such that for every $n \geq 1$,*

$$\sup_t \left| \mathbb{P}_P \left\{ \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \leq t \right\} - \Phi(t) \right| \leq \frac{C\rho}{\sigma^3 \sqrt{n}}.$$

The theorem above concerns the known-variance root. Replacing σ by S_n requires a separate studentized result. Large standardized moments can slow normal approximation. To construct confidence intervals, we can estimate finite-sample features of the sampling distribution, such as skewness, from the data. *The bootstrap approximates this distribution by resampling from the empirical distribution.*

3 Bootstrap sampling distributions

3.1 The empirical bootstrap and conditional notation

Definition 3.1 (Nonparametric bootstrap). Given $\mathcal{D}_n$, draw $X_1^*, \dots, X_n^*$ independently from P_n, with replacement. Compute $\widehat{\theta}_n^*$ by the same rule as $\widehat{\theta}_n$, and put

$$R_n^* = \sqrt{n}(\widehat{\theta}_n^* - \widehat{\theta}_n), \qquad G_n^*(t) = \mathbb{P}(R_n^* \leq t \mid \mathcal{D}_n).$$

We abbreviate $\mathbb{P}^*(\cdot) = \mathbb{P}(\cdot \mid \mathcal{D}_n)$ and $\mathbb{E}^*(\cdot) = \mathbb{E}(\cdot \mid \mathcal{D}_n)$ for bootstrap randomness. These symbols always condition on the original observations.

For a plug-in estimator, the bootstrap treats P_n as the population and P_n^* as the empirical distribution of a new sample. Thus, the bootstrap error measures deviation from $\theta(P_n)$, just as the original sampling error measures deviation from $\theta(P)$. Resampling without replacement with sample size n merely reorders the data and cannot reproduce the variability of a symmetric statistic.

Example 3.2 (Exact bootstrap moments of a mean). For $S_n^2 = P_n(X - \bar{X})^2$,

$$\mathbb{E}^*(\bar{X}^* \mid \mathcal{D}_n) = \bar{X}, \qquad \mathrm{Var}(\bar{X}^* \mid \mathcal{D}_n) = \frac{S_n^2}{n}.$$

Thus $\mathbb{E}^*(R_n^* \mid \mathcal{D}_n) = 0$ and $\mathrm{Var}(R_n^* \mid \mathcal{D}_n) = S_n^2$. The divisor is n because each observation has probability $1/n$ under P_n, and S_n^2 is the exact variance of this discrete distribution. They do not assert that S_n^2 is an unbiased estimator of $\mathrm{Var}_P(X)$.

3.2 The algorithm and its two sources of error

For $b = 1, \dots, B$, independently generate a bootstrap sample given $\mathcal{D}_n$, compute its estimate $\widehat{\theta}_{n,b}^*$, and record $r_b^* = \sqrt{n}(\widehat{\theta}_{n,b}^* - \widehat{\theta}_n)$. The Monte Carlo cdf is

$$\widehat{G}_{n,B}^*(t) = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}\{r_b^* \leq t\}.$$

There are two distinct errors:

$$\|\widehat{G}^*_{n,B} - G_n\|_\infty \le \underbrace{\|\widehat{G}^*_{n,B} - G^*_n\|_\infty}_{\text{Monte Carlo error}} + \underbrace{\|G^*_n - G_n\|_\infty}_{\text{statistical approximation error}}.$$

Conditionally, the DKW inequality from Week 10 yields

$$\mathbb{P}\big(\|\widehat{G}^*_{n,B} - G^*_n\|_\infty > \varepsilon \mid \mathcal{D}_n\big) \le 2e^{-2B\varepsilon^2}.$$

Increasing B reduces simulation error, however it cannot fix a wrong resampling law. At one fixed threshold, the conditional standard deviation is $\{G^*_n(t)(1-G^*_n(t))/B\}^{1/2}$. For example, $B = 9,999$ gives about 0.00218 at a tail probability of 0.05. A uniform cdf bound is more demanding than precision at one threshold.

Bias and standard error summaries. If the required conditional moments exist, write

$$\widehat{\mathrm{bias}}_* = \mathbb{E}(\widehat{\theta}^*_n \mid \mathcal{D}_n) - \widehat{\theta}_n, \qquad \widehat{\mathrm{se}}_* = \{\mathrm{Var}(\widehat{\theta}^*_n \mid \mathcal{D}_n)\}^{1/2}.$$

For stored estimates $\widehat{\theta}^*_{n,1}, \dots, \widehat{\theta}^*_{n,B}$ and their average $\bar{\theta}^*_B$, for $B \ge 2$, the Monte Carlo versions are

$$\widehat{b}_B = \bar{\theta}^*_B - \widehat{\theta}_n, \qquad \widehat{\mathrm{se}}_B = \left\{ \frac{1}{B-1} \sum_{b=1}^{B} (\widehat{\theta}^*_{n,b} - \bar{\theta}^*_B)^2 \right\}^{1/2}.$$

The bias-corrected point estimate is $\widehat{\theta}_n - \widehat{b}_B$. Bias correction can increase variance. Approximating the sampling distribution does not by itself guarantee accurate bias or standard error estimates; these also require control of extreme values. For the sample mean the ideal bootstrap bias is exactly zero and the ideal standard error is $S_n/\sqrt{n}$.

Remark 3.3 (Distributional convergence does not ensure moment convergence). Let $Y_n = n$ with probability $1/n$ and $Y_n = 0$ otherwise. For each fixed $\varepsilon > 0$ and $n > \varepsilon$,

$$\mathbb{P}(|Y_n| > \varepsilon) = \frac{1}{n} \to 0, \qquad \mathbb{E}Y_n = 1, \qquad \mathrm{Var}(Y_n) = n - 1.$$

Thus $Y_n \xrightarrow{\mathbb{P}} 0$, and hence $Y_n \rightsquigarrow 0$, but its mean does not converge to zero and its variance diverges. The rare value n has vanishing probability but a large effect on moments. Similarly, bootstrap distributional consistency alone does not establish consistency of bias or standard error estimates; additional tail control is needed.

3.3 What bootstrap consistency means

Definition 3.4 (Distributional consistency)**.** For scalar roots with a continuous limiting cdf, bootstrap consistency means

$$\Delta_n = \sup_{t \in \mathbb{R}} |\mathbb{P}(R^*_n \le t \mid \mathcal{D}_n) - \mathbb{P}_P(R_n \le t)| \xrightarrow{\mathbb{P}} 0,$$

and thus the convergence is with respect to the randomness of the original sample.

A convenient route is to show that both cdfs converge uniformly to the same continuous G. Weak convergence to a continuous cdf implies uniform convergence of cdfs (Why?). Appendix D gives the finite-grid argument, and also its conditional (bootstrap) form.

Theorem 3.5 (Bootstrap CLT for the sample mean)**.** *Let X_i be iid with mean μ and $0 < \sigma^2 = \mathrm{Var}_P(X) < \infty$. Then*

$$\sup_t \left|\mathbb{P}\{\sqrt{n}(\bar{X}^* - \bar{X}) \le t \mid \mathcal{D}_n\} - \Phi(t/\sigma)\right| \xrightarrow{\mathbb{P}} 0,$$

and consequently the bootstrap is consistent for $\sqrt{n}(\bar{X} - \mu)$. With $S_n^{*2} = n^{-1}\sum_i (X_i^* - \bar{X}^*)^2$, it is also consistent for

$$T_n = \frac{\sqrt{n}(\bar{X} - \mu)}{S_n}, \qquad T_n^* = \frac{\sqrt{n}(\bar{X}^* - \bar{X})}{S_n^*},$$

and $\sup_t |\mathbb{P}(T_n^* \le t \mid \mathcal{D}_n) - \Phi(t)| \xrightarrow{\mathbb{P}} 0$, where we define each root as zero when its denominator vanishes; these events have vanishing probability.

Proof of Theorem 3.5. We verify the conditional CLT conditions, apply the theorem, and use continuity of the Gaussian cdf to obtain uniform convergence.

Step 1: Conditional moments and the CLT conditions. Let $Y_{ni}^* = X_i^* - \bar{X}$. Given $\mathcal{D}_n$, these variables are independent, centered, and have variance S_n^2. The LLN gives $S_n^2 \xrightarrow{\mathbb{P}} \sigma^2 > 0$. We need the following conditional Lindeberg–Feller CLT.

Theorem 3.6 (Conditional Lindeberg–Feller CLT)**.** *Given $\mathcal{D}_n$, let $U_{n1}^*, \ldots, U_{nk_n}^*$ be independent, centered real random variables with finite conditional variances. Put $s_n^2 = \sum_{i=1}^{k_n} \mathbb{E}[(U_{ni}^*)^2 \mid \mathcal{D}_n]$. If $s_n^2 \xrightarrow{\mathbb{P}} v^2$ for a constant $v > 0$ and, for every $\varepsilon > 0$,*

$$L_n(\varepsilon) = \frac{1}{s_n^2}\sum_{i=1}^{k_n} \mathbb{E}\big[(U_{ni}^*)^2 \mathbf{1}\{|U_{ni}^*| > \varepsilon s_n\} \mid \mathcal{D}_n\big] \xrightarrow{\mathbb{P}} 0,$$

where we set $L_n(\varepsilon) = 0$ when $s_n = 0$, then

$$\mathcal{L}\left(\sum_{i=1}^{k_n} U_{ni}^* \,\middle|\, \mathcal{D}_n\right) \rightsquigarrow N(0, v^2) \quad \textit{in } P\textit{-probability}.$$

To obtain the desired result, we apply Theorem 3.6 with $U_{ni}^* = Y_{ni}^*/\sqrt{n}$, $k_n = n$, and $s_n^2 = S_n^2$. It remains to verify its Lindeberg condition: $H_n(\varepsilon)/S_n^2 \xrightarrow{\mathbb{P}} 0$ for every $\varepsilon > 0$, where

$$H_n(\varepsilon) = P_n\big[(X - \bar{X})^2 \mathbf{1}\{|X - \bar{X}| > \varepsilon\sqrt{n} S_n\}\big],$$

where the ratio may be set to zero on $\{S_n = 0\}$, whose probability vanishes.

Step 2: Verify the Lindeberg tail condition. Fix $\varepsilon > 0$. Define $E_n = \{S_n \ge \sigma/2,\ |\bar{X} - \mu| \le 1\}$. The LLN gives $\mathbb{P}_P(E_n) \to 1$. On E_n, whenever $|X - \bar{X}| > \varepsilon\sqrt{n} S_n$, the triangle inequality gives

$$|X - \mu| \ge |X - \bar{X}| - |\bar{X} - \mu| > \frac{\varepsilon\sigma}{2}\sqrt{n} - 1 \ge \frac{\varepsilon\sigma}{4}\sqrt{n}$$

for all sufficiently large n. Thus, with $c = \varepsilon\sigma/4$, on E_n we have

$$\mathbf{1}\{|X - \bar{X}| > \varepsilon\sqrt{n}S_n\} \le \mathbf{1}\{|X - \mu| > c\sqrt{n}\}.$$

(When the first indicator is zero, the inequality is automatic.) Using $(u+v)^2 \le 2u^2 + 2v^2$, we obtain, on E_n,

$$H_n(\varepsilon) \le 2P_n[(X-\mu)^2\mathbf{1}\{|X-\mu| > c\sqrt{n}\}] + 2(\bar{X}-\mu)^2.$$

The empirical tail term has expectation

$$\mathbb{E}_P\big[P_n\{(X-\mu)^2\mathbf{1}(|X-\mu| > c\sqrt{n})\}\big] = \mathbb{E}_P[(X-\mu)^2\mathbf{1}\{|X-\mu| > c\sqrt{n}\}] \to 0,$$

by integrability of $(X-\mu)^2$. Markov's inequality makes this term $o_P(1)$, and the LLN gives $(\bar{X} - \mu)^2 = o_P(1)$. Since the bound holds on E_n and $\mathbb{P}_P(E_n^c) \to 0$, we obtain $H_n(\varepsilon) \xrightarrow{\mathbb{P}} 0$. Together with $S_n^2 \xrightarrow{\mathbb{P}} \sigma^2 > 0$, this gives $H_n(\varepsilon)/S_n^2 \xrightarrow{\mathbb{P}} 0$, as required.

Step 3: Conclude the CLT and bootstrap consistency. Theorem 3.6 therefore implies

$$\mathcal{L}\{\sqrt{n}(\bar{X}^* - \bar{X}) \mid \mathcal{D}_n\} \rightsquigarrow N(0, \sigma^2) \quad \text{in } P\text{-probability}.$$

To pass from conditional weak convergence to uniform cdf convergence, we use the following lemma.

Lemma 3.7 (Continuous limits give uniform cdf convergence)**.** *If cdfs H_n converge pointwise to a continuous cdf H, then $\|H_n - H\|_\infty \to 0$. If H_n are random cdfs and pointwise convergence is in probability, the uniform convergence is also in probability.*

Its proof is given in Appendix D.1. Since the Gaussian cdf is continuous, Lemma 3.7 upgrades this to the uniform cdf convergence stated in the theorem. The original mean CLT gives the same uniform cdf limit $\Phi(t/\sigma)$. The triangle inequality therefore proves bootstrap consistency for the unstudentized root.

The proof of the studentized statements is left as Exercise 1. □

Remark 3.8 (Intuition). *The bootstrap mean fluctuates around the observed mean. Its empirical variance converges to the population variance.* The same Gaussian fluctuation therefore appears in both sampling worlds. This result needs only second moments; higher-order claims below need much stronger assumptions.

Theorem 3.9 (Bootstrap for smooth functions of means)**.** *Under the assumptions of Theorem 2.2, resample the vectors Z_i jointly and set $\widehat{\theta}_n^* = h(\bar{Z}_n^*)$. Then*

$$\sup_t \left|\mathbb{P}\{\sqrt{n}(\widehat{\theta}_n^* - \widehat{\theta}_n) \le t \mid \mathcal{D}_n\} - \Phi(t/\sigma)\right| \xrightarrow{\mathbb{P}} 0.$$

With the sample-gradient covariance estimate from Theorem 2.2, recomputed in every bootstrap sample, the corresponding studentized bootstrap root converges conditionally to $N(0,1)$.

The proof is given in Appendix B.2. The notation $o_{P^*}(1)$ in P-probability means that, for every $\varepsilon, \delta > 0$, $\mathbb{P}_P\{\mathbb{P}(|r_n^*| > \varepsilon \mid \mathcal{D}_n) > \delta\} \to 0$. A conditional $O_{P^*}(1)$ has the analogous tightness meaning.

3.4 Beyond smooth means: what must be checked

To extend the bootstrap argument beyond smooth functions of means, we establish corresponding linear expansions for the original and bootstrap estimators, using the same influence function:

$$\sqrt{n}(\widehat{\theta}_n - \theta) = n^{-1/2} \sum_i \varphi_P(X_i) + o_P(1),$$
$$\sqrt{n}(\widehat{\theta}^*_n - \widehat{\theta}_n) = n^{-1/2} \sum_i \{\varphi_P(X^*_i) - P_n\varphi_P\} + o_{P^*}(1),$$

with the second remainder vanishing conditionally in P-probability. Once both expansions have been established, applying Theorem 3.5 to $\varphi_P(X_i)$ gives the conditional Gaussian limit of the bootstrap linear term. The negligible remainders then imply that the original and bootstrap roots have the same Gaussian limit. Lemma 3.7 and the triangle inequality yield bootstrap distributional consistency. Establishing the second expansion is a substantive step; it can fail for nonsmooth or boundary problems (see Section 5.2 for an endpoint example of bootstrap failure).

For the sample median m, positive continuous density $f(m)$ gives the first-order influence function $\varphi_P(x) = \{1/2 - \mathbf{1}(x \le m)\}/f(m)$. A bootstrap result also needs a conditional quantile expansion. The smooth-means theorem does not cover the median directly. Dependence likewise requires a resampling scheme that preserves the relevant dependence, such as blocks under additional assumptions.

Distributional convergence also does not imply convergence of variances or expectations. Consistency of a bootstrap standard error for a general statistic needs suitable moment control, often uniform integrability. For the mean, the exact identity in Example 3.2 supplies this control directly.

4 From bootstrap distributions to inference

4.1 Basic and percentile intervals

For any cdf H, define its generalized p-quantile by $H^{-1}(p) = \inf\{t : H(t) \ge p\}$. Let q^*_p be the conditional p-quantile of $R^*_n = \sqrt{n}(\widehat{\theta}^*_n - \widehat{\theta}_n)$. Inverting $q^*_{\alpha/2} \le \sqrt{n}(\widehat{\theta}_n - \theta) \le q^*_{1-\alpha/2}$ gives the *basic bootstrap interval*

$$I_{\text{basic}} = \left[\widehat{\theta}_n - \frac{q^*_{1-\alpha/2}}{\sqrt{n}}, \quad \widehat{\theta}_n - \frac{q^*_{\alpha/2}}{\sqrt{n}}\right].$$

The upper error quantile determines the lower parameter endpoint. This sign reversal matters when the error distribution is asymmetric.

If a^*_p denotes the conditional p-quantile of $\widehat{\theta}^*_n$ itself, then $a^*_p = \widehat{\theta}_n + q^*_p/\sqrt{n}$. The *percentile interval* is $[a^*_{\alpha/2}, a^*_{1-\alpha/2}]$. It does not make the same inversion as the basic interval. Under the centered normal limits proved above, both are first-order valid because the limiting error distribution is symmetric. Bootstrap consistency for a general asymmetric root does not, by itself, justify the percentile interval.

4.2 Bootstrap-t intervals

Let c^*_p be the conditional p-quantile of

$$T^*_n = \frac{\sqrt{n}(\widehat{\theta}^*_n - \widehat{\theta}_n)}{\widehat{\sigma}^*_n}, \qquad \widehat{\sigma}^*_n \text{ recomputed from the bootstrap sample.}$$

The bootstrap-t interval is

$$I_t = \left[\widehat{\theta}_n - c^*_{1-\alpha/2}\frac{\widehat{\sigma}_n}{\sqrt{n}},\quad \widehat{\theta}_n - c^*_{\alpha/2}\frac{\widehat{\sigma}_n}{\sqrt{n}}\right].$$

The letter t means studentization; the conditional reference need not be a Student distribution.

Proposition 4.1 (First-order quantile inversion)**.** *Suppose the original and conditional bootstrap roots both converge to a continuous cdf G, and G is strictly increasing near its $\alpha/2$ and $1-\alpha/2$ quantiles. Then the basic interval has limiting coverage $1-\alpha$. If these conditions hold for the studentized roots, the bootstrap-t interval has the same limiting coverage.*

Proof of Proposition 4.1. Uniform conditional cdf convergence and local strict increase imply $q^*_p \xrightarrow{\mathbb{P}} G^{-1}(p)$ at either required p. To see this, choose $\varepsilon > 0$: the limiting cdf is below p at $G^{-1}(p) - \varepsilon$ and above p at $G^{-1}(p) + \varepsilon$, for sufficiently small ε. Uniform convergence traps the bootstrap quantile between these two values with probability tending to one. Slutsky applies jointly to the original root and these probability-convergent cutoffs, even though they use the same observations. Continuity of G gives probability $1-\alpha$ between the limiting cutoffs. The studentized case is identical after inversion with its estimated scale. □

Example 4.2 (An asymmetric pair of error quantiles)**.** Suppose $\widehat{\theta}_n = 4$, $\sqrt{n} = 10$, and the bootstrap error quantiles are $q^*_{.025} = -1$, $q^*_{.975} = 2$. The basic interval is $[3.8, 4.1]$; the percentile interval is $[3.9, 4.2]$. They are different even with infinitely many bootstrap replications.

4.3 Bootstrap tests and fitted models

For a one-sided null $H_0 : \theta = \theta_0$, a studentized test compares $T_{n,0} = \sqrt{n}(\widehat{\theta}_n - \theta_0)/\widehat{\sigma}_n$ with a conditional quantile of the centered bootstrap root. The preceding consistency result gives pointwise asymptotic level under the null. An unrestricted resample followed by the raw, uncentered statistic $\widehat{\theta}^*_n - \theta_0$ generally retains the observed shift and does not estimate the null error law.

A *parametric bootstrap* instead draws from a fitted model $P_{\widehat{\eta}}$, with parameter η estimated from the data. *For a fitted goodness-of-fit statistic from Week 10, one simulates from the fitted null model, refits η in each simulated sample, and recomputes the fitted discrepancy.* This reproduces the fitting step. Validity requires model-specific regularity and conditional convergence of the fitted statistic; refitting is necessary but is not a universal validity theorem. Model misspecification is also distinct from Monte Carlo error.

5 Subsampling and nonstandard limits

5.1 Use smaller samples and the correct rate

Suppose $\widehat{\theta}_n = T_n(X_1, \ldots, X_n)$ is a symmetric estimator and, for a known deterministic rate $\tau_n \to \infty$,

$$\tau_n(\widehat{\theta}_n - \theta) \rightsquigarrow J,$$

where we also use J for the limiting cdf. For a subset $I \subset \{1, \ldots, n\}$ of size b, let $\widehat{\theta}_{b,I}$ be the same estimator computed only on that subset. The all-subsets cdf is

$$L_{n,b}(t) = \binom{n}{b}^{-1} \sum_{|I|=b} \mathbf{1}\{\tau_b(\widehat{\theta}_{b,I} - \widehat{\theta}_n) \le t\}.$$

Subsampling draws observations *without replacement within each subset*. *We use τ_b to construct the subsample root, then use τ_n when inverting its quantiles for the full-sample parameter.*

Theorem 5.1 (Iid subsampling consistency)**.** *Assume iid observations, a symmetric estimator, and a continuous limiting cdf J as above. Choose $b = b_n$ such that*

$$b \to \infty, \qquad b/n \to 0, \qquad \tau_b/\tau_n \to 0.$$

Then $\sup_t |L_{n,b}(t) - J(t)| \xrightarrow{\mathbb{P}} 0$. Thus its quantiles give first-order valid basic intervals wherever J is locally strictly increasing.

The first condition lets the subsample reach its own limiting law. The second leaves enough effectively distinct subsets. The third makes replacement of θ by $\widehat{\theta}_n$ negligible at the subsample scale. For $\tau_n = n^a$ with $a > 0$, $b/n \to 0$ implies the third condition. For an arbitrary rate, it must be checked separately. The proof uses bounded symmetric averages and appears in Appendix C.

If ℓ_p denotes a quantile of $L_{n,b}$, the interval is

$$\left[\widehat{\theta}_n - \ell_{1-\alpha/2}/\tau_n, \quad \widehat{\theta}_n - \ell_{\alpha/2}/\tau_n\right].$$

One can approximate $L_{n,b}$ by drawing B independent uniform subsets conditionally on the data. Subsets can overlap across draws. The conditional Monte Carlo cdf has the same DKW bound as before. Overlap does not make the all-subsets indicators independent under the original sampling law; the proof below accounts for their dependence.

5.2 A worked non-Gaussian example: an upper endpoint

Let $X_i \overset{\text{iid}}{\sim} \text{Unif}[0, \theta]$, where $\theta > 0$, and let $M_n = \max_i X_i$. For $y \geq 0$ and sufficiently large n,

$$\mathbb{P}_P\{n(\theta - M_n) > y\} = \left(1 - \frac{y}{n\theta}\right)^n \longrightarrow e^{-y/\theta}.$$

Thus $n(M_n - \theta)$ converges to a negative exponential random variable, with continuous cdf

$$J(t) = \begin{cases} e^{t/\theta}, & t \leq 0, \\ 1, & t > 0. \end{cases}$$

The correct rate is n, not $\sqrt{n}$.

The ordinary bootstrap maximum M_n^* cannot exceed M_n. For continuous data, the sample maximum is unique almost surely, and

$$\mathbb{P}\{n(M_n^* - M_n) = 0 \mid \mathcal{D}_n\} = 1 - (1 - 1/n)^n \longrightarrow 1 - e^{-1}.$$

This atom persists, whereas J has no atom at zero. More explicitly, a jump of size a in one cdf forces its uniform distance from any continuous cdf to be at least $a/2$. Ordinary bootstrap distributional consistency therefore fails in this example.

Subsampling uses $b(M_{b,I} - M_n)$. Theorem 5.1 applies when $b \to \infty$ and $b/n \to 0$, because $\tau_b/\tau_n = b/n$. In particular, with $\ell_\alpha \approx \theta \log \alpha < 0$, an asymptotic one-sided upper confidence bound is $M_n - \ell_\alpha/n$. The exact parametric upper bound is $M_n/\alpha^{1/n}$, obtained from the exact law of M_n/θ. Its expansion agrees with the subsampling bound to first order when ℓ_α consistently estimates $\theta \log \alpha$.

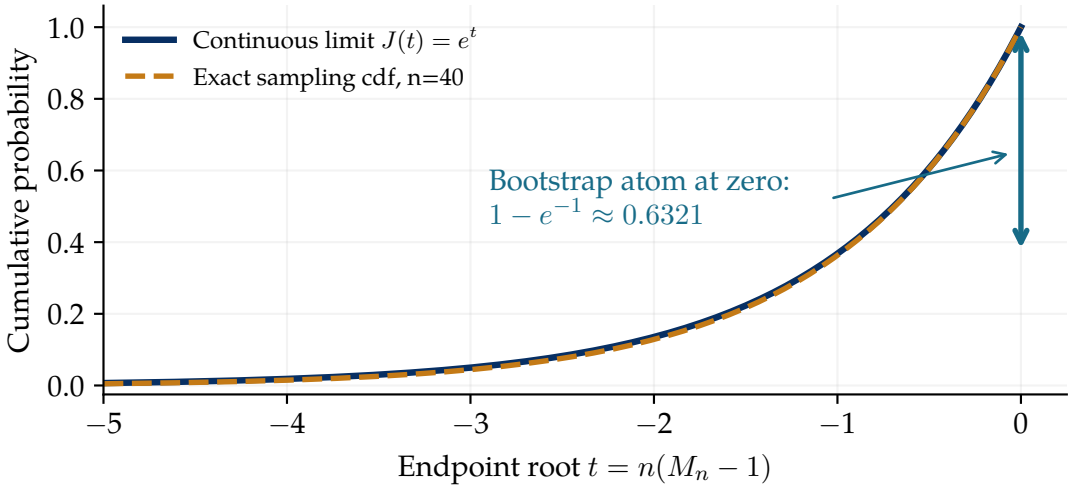


Figure 1: Endpoint example at $\theta = 1$. The limiting cdf of $n(M_n - 1)$ is continuous. An ordinary bootstrap root has a persistent atom at zero; the arrow marks its asymptotic size. The orange curve is the exact finite-n sampling cdf, not a simulated bootstrap cdf.

5.3 Choosing a subsample size; the m-out-of-n bootstrap

Theory gives a range of admissible b, not an optimal finite-sample value. Small b increases the gap between subsample and limiting laws. Large b/n makes centering and subset dependence more important. Compare several sizes, for example $b = \lfloor n^\kappa \rfloor$ for fixed $0 < \kappa < 1$, while keeping the rate and parameter fixed. Sensitivity analysis does not itself prove coverage.

An m-out-of-n bootstrap draws m observations *with replacement* from P_n and uses $\tau_m(\widehat{\theta}^*_m - \widehat{\theta}_n)$. It differs from subsampling. Appropriate $m \to \infty$, $m/n \to 0$ conditions can repair some nonregular problems, but the subsampling theorem does not automatically prove this method valid. In dependent data, arbitrary subsets are generally inappropriate; block subsampling needs its own assumptions.

6 Permutation and randomization tests

6.1 Permutation tests use a null symmetry

Let W denote the full observed data and let $\mathcal{G}$ be a finite group of transformations, including the identity. A group is closed under composition and inverses. *Under the null, assume gW has the same distribution as W for every $g \in \mathcal{G}$.* This is a *null invariance* assumption. For a statistic $A(W)$ with larger values more extreme, define

$$p_{\text{perm}}(W) = \frac{1}{|\mathcal{G}|} \sum_{g \in \mathcal{G}} \mathbf{1}\{A(gW) \geq A(W)\}.$$

The observed arrangement is included. Ties are counted in the upper tail.

Theorem 6.1 (Finite-sample permutation validity)**.** *Under null invariance, $\mathbb{P}_P\{p_{\text{perm}}(W) \leq \alpha\} \leq \alpha$ for every $\alpha \in [0, 1]$. No large-sample approximation is required.*

Proof of Theorem 6.1. Fix w. The multiset $\{A(gw) : g \in \mathcal{G}\}$ is unchanged by replacing w by hw, since $g \mapsto gh$ is a bijection. Among these $|\mathcal{G}|$ values, at most an α fraction have upper-tail rank, with ties included, at most $\alpha|\mathcal{G}|$. Thus $|\mathcal{G}|^{-1}\sum_h \mathbf{1}\{p_{\text{perm}}(hw) \le \alpha\} \le \alpha$. Null invariance makes the expectation of each summand equal to $\mathbb{P}_P\{p_{\text{perm}}(W) \le \alpha\}$. Taking expectations proves the result. □

The conclusion is exact finite-sample *level control*. Because of discreteness or ties, the rejection probability can be below α. Randomization at a boundary can obtain equality; this is separate from random assignment of treatment.

6.2 Two samples: equality of distributions and equality of means

For independent samples $X_1, \dots, X_r \overset{\text{iid}}{\sim} P$ and $Y_1, \dots, Y_s \overset{\text{iid}}{\sim} Q$, the null $P = Q$ makes the pooled observations exchangeable: their joint distribution is unchanged by reordering. Permute the group labels while preserving group sizes, recompute A, and use the rank formula. This also calibrates the two-sample KS statistic from Week 10, including ties, under the common-distribution null.

The weaker null $\mathbb{E}_P X = \mathbb{E}_Q Y$ permits different variances and does not imply exchangeability. For $N = r+s$, $r/N \to \lambda \in (0,1)$, and common means, the actual asymptotic variance of $\sqrt{N}(\bar{X}-\bar{Y})$ is

$$V_{\text{actual}} = \frac{\sigma_P^2}{\lambda} + \frac{\sigma_Q^2}{1-\lambda}.$$

The permutation reference for the raw mean difference instead has limiting variance

$$V_{\text{perm}} = \frac{\lambda\sigma_P^2 + (1-\lambda)\sigma_Q^2}{\lambda(1-\lambda)}.$$

These usually differ for unequal variances and unequal sample fractions. Equality of means alone does not justify the raw-label permutation test.

A useful repair is to permute and recompute the Welch statistic

$$A_{\text{W}} = \frac{\bar{X} - \bar{Y}}{\sqrt{S_X^2/r + S_Y^2/s}},$$

where the within-group variances are recomputed after every relabeling. For example, assume $r/N \to \lambda \in (0,1)$, positive finite group variances, and finite $(2+\delta)$th absolute moments in both groups for some $\delta > 0$. Under the common-mean null, studentization then makes both reference laws asymptotically standard normal. It retains exact permutation validity under $P = Q$. This is an advanced asymptotic result; see Chung and Romano [5] for assumptions and proof. Finite-sample exactness under the weaker mean null is not claimed.

6.3 Randomization tests use the assignment design

In a completely randomized experiment with N units, exactly r units receive treatment. Let $Z_i \in \{0,1\}$ denote assignment and let $Y_i(1), Y_i(0)$ denote fixed potential outcomes. The observed outcome is $Y_i^{\text{obs}} = Z_i Y_i(1) + (1-Z_i)Y_i(0)$. Assume the assignment is uniform over the $\binom{N}{r}$ allowed vectors.

Under Fisher's *sharp null*, $Y_i(1) = Y_i(0)$ for every unit, so every missing potential outcome is known. *Hold this outcome vector fixed, draw assignments from the actual design, and recompute a statistic,*

such as the treated-minus-control mean. Its design-based upper-tail rank is a valid p-value conditional on the fixed potential outcomes. The same weighted-tail rank argument applies to a nonuniform known design using its assignment probabilities. Uniform relabeling is appropriate only for a uniform design; paired and blocked experiments require design-respecting assignments.

The weak null $N^{-1}\sum_i\{Y_i(1) - Y_i(0)\} = 0$ does not impute all potential outcomes. The sharp-null argument therefore does not establish exact validity for a zero average effect. Observational exchangeability and experimental random assignment are different assumptions, even when the resulting computations look similar.

Example 6.2 (Four units and six assignments)**.** Under a sharp null, fix outcomes $(1, 2, 4, 7)$ and assign two units to treatment. The six treated-minus-control mean differences are

$$-4,\ -2,\ 1,\ -1,\ 2,\ 4.$$

If units with outcomes 4 and 7 were treated, the observed difference is 4. The upper-tail p-value is $1/6$; using the absolute difference gives $2/6$. These fractions include the observed assignment.

6.4 Monte Carlo permutation p-values

If enumeration is costly, sample B independent uniform group transformations, independently of the data. With $A_0 = A(W)$ and $A_b = A(g_bW)$, use

$$p_B = \frac{1 + \sum_{b=1}^{B} \mathbf{1}\{A_b \geq A_0\}}{B + 1}.$$

Under null invariance, the observed statistic and the transformed statistics have exchangeable ranks; the formula gives finite-sample level control jointly over the data and transformation draws. Its minimum is $1/(B + 1)$. The same construction applies to iid assignments from the known design under a sharp null. For a two-sided test, choose A to measure two-sided extremeness before resampling.

The extra one is justified by rank validity, not by arbitrary numerical smoothing. The raw fraction $B^{-1}\sum_b \mathbf{1}\{A_b \geq A_0\}$ estimates a conditional tail probability, but can be zero and need not be a valid finite-B p-value. Sampling transformations without replacement, omitting the identity, or using restricted transformations requires checking the corresponding rank construction; see Phipson and Smyth [7].

7 Higher-order correctness

7.1 First-order validity and second-order refinement

First-order consistency says $\Delta_n \xrightarrow{\mathbb{P}} 0$. It does not state how fast. For a studentized statistic in a smooth setting, a normal approximation often has cdf error of order $n^{-1/2}$. We call a bootstrap approximation *second-order correct* when its cdf error is $o_P(n^{-1/2})$. Under stronger conditions an $O_P(n^{-1})$ error is possible. These are distributional error statements. Coverage error of a confidence interval is a separate quantity.

An *Edgeworth expansion* adds correction terms to a Gaussian cdf using cumulants, which describe features such as skewness. For the known-variance standardized mean $Z_n = \sqrt{n}(\bar{X} - \mu)/\sigma$, the

first correction has the form

$$\mathbb{P}_P(Z_n \le x) = \Phi(x) + \frac{\gamma}{6\sqrt{n}}(1 - x^2)\phi(x) + \text{remainder}.$$

For the studentized mean $T_n = \sqrt{n}(\bar{X} - \mu)/S_n$ with the divisor-n variance, it instead has the form

$$\mathbb{P}_P(T_n \le x) = \Phi(x) + \frac{\gamma}{6\sqrt{n}}(2x^2 + 1)\phi(x) + \text{remainder}. \tag{7.1}$$

Studentization changes the correction polynomial. An Edgeworth approximation is an asymptotic expansion; its truncation need not be a cdf in extreme tails.

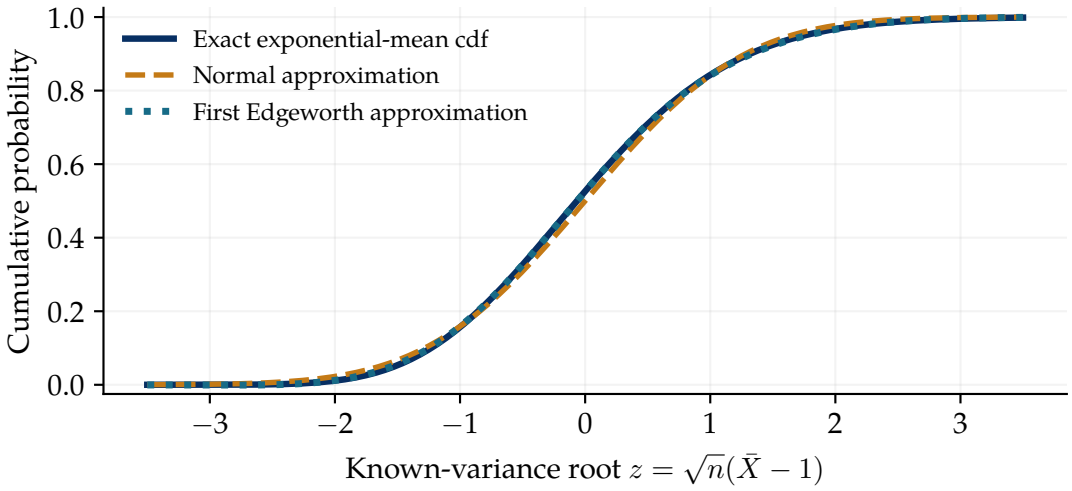


Figure 2: Exact distribution of the known-variance mean root $Z_n = \sqrt{n}(\bar{X} - 1)$ for exponential data, $n = 25$, compared with normal and first Edgeworth approximations. The exact curve uses the gamma distribution; no simulation is used. This figure illustrates the correction mechanism, not the studentized expansion in (7.1).

7.2 An explicit theorem for matching expansions

To keep the logical requirements visible, we state a conditional theorem. Its expansion assumptions isolate the analytic work needed before claiming a rate. Such expansions are available for sufficiently regular smooth statistics with adequate moments and characteristic-function conditions; they do not follow from the finite-variance bootstrap CLT. Hall [3], Chapters 2–3 and 5, develops these conditions and their proofs.

For the scalar mean define

$$\widehat{\gamma}_n = \frac{n^{-1}\sum_i (X_i - \bar{X})^3}{S_n^3}, \qquad A(x) = \frac{2x^2 + 1}{6}\phi(x).$$

Assumption 7.1 (Uniform matched expansions)**.** The variance is positive, $\mathbb{E}|X|^6 < \infty$, and the

original and ideal conditional studentized cdfs admit

$$\mathbb{P}_P(T_n \le x) = \Phi(x) + n^{-1/2}\gamma A(x) + r_n(x),$$
$$\mathbb{P}(T_n^* \le x \mid \mathcal{D}_n) = \Phi(x) + n^{-1/2}\widehat{\gamma}_n A(x) + r_n^*(x),$$

where $\|r_n\|_\infty = O(n^{-1})$ and $\|r_n^*\|_\infty = O_P(n^{-1})$.

The moment condition controls empirical skewness. The two expansion and remainder claims are additional assumptions, not consequences of the sixth moment alone. For a general statistic the correction has several estimated cumulant coefficients, rather than just skewness.

A characteristic function is $\chi_V(t) = \mathbb{E}e^{it^{\mathrm{T}}V}$. Conditions on it control oscillation away from zero. A standard form is Cramér's condition,

$$\limsup_{\|t\|\to\infty} |\chi_V(t)| < 1,$$

for the vector V used in the expansion, often containing both X and X^2 for studentization. Lattice distributions require adjusted expansions. We do not infer a conditional expansion by applying this condition directly to a discrete empirical distribution; that bootstrap analytic step needs its own argument.

Theorem 7.2 (Matching the first correction)**.** *Under Assumption 7.1,*

$$\sup_x |\mathbb{P}(T_n^* \le x \mid \mathcal{D}_n) - \mathbb{P}_P(T_n \le x)| = O_P(n^{-1}).$$

If $\gamma \neq 0$, the error of the standard normal cdf is of order $n^{-1/2}$ under the same original expansion.

Proof of Theorem 7.2. Finite sixth moment gives a multivariate root-n CLT for the empirical first, second, and third raw moments. Skewness is a continuously differentiable function of these moments where variance is positive. The delta method gives $\widehat{\gamma}_n - \gamma = O_P(n^{-1/2})$. Write H_n and H_n^* for the original and conditional studentized cdfs. Subtract the expansions to obtain

$$\|H_n^* - H_n\|_\infty \le n^{-1/2}|\widehat{\gamma}_n - \gamma|\|A\|_\infty + \|r_n^*\|_\infty + \|r_n\|_\infty = O_P(n^{-1}).$$

The function A is bounded. If $\gamma \neq 0$, evaluation at $x = 0$ gives a nonzero $n^{-1/2}$ leading normal-approximation error, while the uniform bound supplies the matching upper order. □

Remark 7.3 (Intuition). The normal limit keeps the leading Gaussian term. *The bootstrap estimates the first correction from the observed sample.* Estimating its coefficient at rate $n^{-1/2}$, then multiplying by its $n^{-1/2}$ prefactor, leaves an error of order n^{-1}.

An unstudentized bootstrap need not have this improvement. Its leading Gaussian law contains the estimated variance: even $\Phi(x/S_n) - \Phi(x/\sigma)$ is generally $O_P(n^{-1/2})$. *Studentization removes this nuisance parameter from the leading law.* If skewness vanishes, the leading normal error may already cancel, so a strict order improvement is not automatic.

7.3 Quantile corrections and coverage: distinct statements

A *Cornish–Fisher correction* approximately inverts an Edgeworth cdf. Put $q_p = z_p + n^{-1/2} d_p$ in (7.1). Taylor expansion gives, for fixed $p \in (0,1)$,

$$d_p = -\frac{\gamma}{6}(2z_p^2 + 1), \qquad q_p = z_p - \frac{\gamma}{6\sqrt{n}}(2z_p^2 + 1) + O(n^{-1}),$$

under the stated uniform expansion and local positive-density regularity. The bootstrap replaces γ by $\widehat{\gamma}_n$ and approximates the full conditional quantile rather than explicitly truncating this formula.

For $p = .95$, $n = 25$, and exponential skewness $\gamma = 2$, this first correction moves $z_{.95} \approx 1.645$ to about 1.217 for the *studentized* root. This is an asymptotic illustration, not an exact small-sample quantile. The correction differs from the known-variance figure because the two roots differ.

For bootstrap-t intervals, classical smooth-model theory can give one-sided coverage error of order n^{-1} under additional regularity and joint expansion conditions [3]. We have proved distributional refinement under Assumption 7.1, and only first-order coverage in Proposition 4.1. A cdf approximation at deterministic thresholds does not alone prove a higher-order coverage rate with data-dependent cutoffs: the root and its estimated cutoff use the same data. Symmetric two-sided intervals can have further cancellations, so their coverage order should not be read directly from a one-sided cdf rate.

With finite B, the conditional Monte Carlo cdf error is $O_{P^*}(B^{-1/2})$. A sufficient order choice for this uniform error to be negligible relative to n^{-1} is $B/n^2 \to \infty$. This is an asymptotic cdf comparison, not a rule that every practical bootstrap needs more than n^2 draws. A fixed B does not support an ideal-bootstrap higher-order limit as $n \to \infty$.

8 A common view of the methods

Method	Reference construction	Main validity requirement
Normal approximation	Limiting Gaussian root	Linearization, CLT, consistent scale
Empirical bootstrap	Iid draws from P_n, with replacement	Conditional law reproduces the original root
Subsampling	Size-b subsets, without replacement	Known rate, continuous limit, $b/n \to 0$, negligible recentering
Permutation test	Null-invariant transformations	Exchangeability or other group invariance
Randomization test	Actual treatment assignments	Known design and an imputable null

For smooth means, bootstrap consistency and normal approximation have the same first-order Gaussian explanation. Studentization can additionally let the bootstrap match a skewness correction. For an endpoint, a smaller sample and the correct rate recover a non-Gaussian law. For permutation or randomization tests, finite-sample validity comes from rank symmetry rather than a CLT.

The practical sequence is to specify the inferential target, define its centered root and scale, identify what is random under the target law, and reproduce that randomness. Refit all components that the original analysis estimates. Distinguish statistical approximation from Monte Carlo approximation, and distinguish pointwise asymptotic validity from finite-sample level control.

Exercises

Exercise 1 (Studentized bootstrap mean)**.** Under the assumptions of Theorem 3.5, define

$$S_n^{*2} = \frac{1}{n}\sum_{i=1}^{n}(X_i^* - \bar{X}^*)^2, \qquad T_n^* = \frac{\sqrt{n}(\bar{X}^* - \bar{X})}{S_n^*}, \qquad T_n = \frac{\sqrt{n}(\bar{X} - \mu)}{S_n}.$$

Set each root to zero when its denominator vanishes. First show that, for every $\eta > 0$, $\mathbb{P}(|S_n^{*2} - \sigma^2| > \eta \mid \mathcal{D}_n) \xrightarrow{\mathbb{P}} 0$. Then prove

$$\sup_t |\mathbb{P}(T_n^* \le t \mid \mathcal{D}_n) - \Phi(t)| \xrightarrow{\mathbb{P}} 0$$

and deduce bootstrap consistency for T_n. Explain why the zero-denominator events have vanishing probability and why no fourth-moment assumption is needed. *Hint.* Apply Lemma D.1 to X and X^2, then use conditional Slutsky (Lemma D.2), Lemma 3.7, and the original studentized CLT.

Exercise 2 (Bootstrap counts)**.** Let N_i^* count selections of observation i in a size-n empirical bootstrap sample. Show that $(N_1^*, \ldots, N_n^*)$ is multinomial with probabilities $1/n$. Derive its covariance matrix and recover Example 3.2 from $\bar{X}^* = n^{-1}\sum_i N_i^* X_i$.

Exercise 3 (A smooth nonlinear statistic)**.** For a positive mean μ and finite positive variance, apply the original and bootstrap delta methods to $\log \bar{X}$. State an event on which the statistic is defined and explain why its probability tends to one. Derive a studentized root and both interval endpoints.

Exercise 4 (Centering and inversion)**.** Use the asymmetric quantiles in Section 4 to verify the basic and percentile intervals. Explain why multiplying a bootstrap error quantile by $\sqrt{n}$, rather than dividing by it, gives the wrong scale.

Exercise 5 (An endpoint)**.** Derive the exact cdf of $n(M_n - \theta)$ and the exact upper confidence bound in Section 5.2. For $b = \lfloor n^{2/3} \rfloor$, check every condition in Theorem 5.1. Explain the bootstrap atom at zero.

Exercise 6 (A weak null is not exchangeability)**.** For $\lambda = 1/3$, $\sigma_P^2 = 1$, and $\sigma_Q^2 = 4$, compute V_{actual} and V_{perm}. Is the raw mean-difference permutation reference too dispersed or too concentrated? Which assumption gives finite-sample exactness?

Exercise 7 (Assignment and Monte Carlo ranks)**.** Enumerate Example 6.2. For $B = 99$ and no sampled transformation as extreme as the observation, compute the corrected p-value. Explain why the raw estimate zero and the corrected p-value answer different questions.

Exercise 8 (The first correction)**.** Invert the two Edgeworth expansions in Section 7. Compare their correction polynomials, and derive the $O_P(n^{-1})$ cdf difference from root-n skewness estimation. Explain why this calculation does not prove a coverage rate for a random cutoff.

Supplementary Material

Overview

Appendix A fixes conditional-convergence notation and proves the conditional Lindeberg–Feller theorem. Appendix B proves the original and bootstrap smooth-means results, including covariance convergence in operator norm. Appendix C proves iid subsampling consistency. Appendix D collects the cdf convergence proof, conditional LLN, and conditional Slutsky lemma. Appendix E records the exact numerical illustrations. The main text contains the conditional bootstrap mean CLT proof, as well as the quantile-inversion, permutation-rank, and matched-expansion proofs.

Guide to proofs of main results

Theorem 2.2 is proved in Appendix B by Taylor expansion and the CLT; fixed dimension converts entrywise covariance convergence into operator-norm convergence. Theorem 3.6 is proved in Appendix A.1 by applying the ordinary Lindeberg–Feller theorem along almost-sure subsequences. The proof of Theorem 3.5 is given in Section 3. It uses the Lindeberg condition for the empirical triangular array and Lemma 3.7 for uniform cdf convergence. Exercise 1 extends the result to studentization using Lemma D.1. Theorem 3.9 is proved in Appendix B.2 by reducing to the conditional mean CLT after Taylor expansion. Theorem 5.1 uses a random-partition bound for a bounded symmetric average (Lemma C.1), then removes a negligible centering shift. Theorem 7.2 assumes the analytic expansions and proves only their algebraic matching consequence; it is not a proof of the expansion assumptions.

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A Preliminaries: conditional limits

All conditional convergence statements are in probability with respect to the original sample. One useful precise formulation is bounded-Lipschitz convergence. A function $f : \mathbb{R}^d \to \mathbb{R}$ is bounded Lipschitz in the unit class if $|f| \le 1$ and $|f(x) - f(y)| \le \|x - y\|$. Conditional weak convergence of V_n^* to V means

$$\sup_{f \text{ in the unit class}} |\mathbb{E}\{f(V_n^*) \mid \mathcal{D}_n\} - \mathbb{E}f(V)| \xrightarrow{\mathbb{P}} 0.$$

In finite dimensions, this is equivalent to conditional weak convergence in probability. Subsequence reasoning is often convenient: every subsequence has a further subsequence along which the empirical moment and Lindeberg conditions hold almost surely. On that further subsequence, ordinary deterministic triangular-array arguments apply for almost every original data sequence. This gives the desired in-probability conditional statement for the full sequence.

A.1 Conditional Lindeberg–Feller theorem

The following proof applies the ordinary triangular-array CLT after fixing the original data along a suitable subsequence.

Proof of Theorem 3.6. From any subsequence, select a further subsequence along which $s_n^2 \to v^2$ and $L_n(1/m) \to 0$ almost surely for every positive integer m. Monotonicity in ε then gives the Lindeberg condition for every $\varepsilon > 0$. For almost every original data sequence, the conditional laws form a deterministic triangular array satisfying the ordinary Lindeberg–Feller theorem. Thus the conditional laws converge weakly to $N(0, v^2)$ along this further subsequence. The subsequence criterion gives conditional weak convergence in P-probability. □

B Proofs for smooth functions of means

B.1 The delta method and variance consistency

The proof separates Euclidean convergence of vectors, operator-norm convergence of covariance matrices, and scalar convergence of the estimated variance.

Proof of Theorem 2.2. Write $\|v\|_2 = (\sum_j v_j^2)^{1/2}$ for the Euclidean norm and $\|A\|_{\mathrm{op}} = \sup_{\|v\|_2=1} \|Av\|_2$ for its induced matrix operator norm. All vector convergence below uses $\|\cdot\|_2$; matrix convergence uses $\|\cdot\|_{\mathrm{op}}$.

Linearization and the limiting distribution. The coordinatewise LLN gives $\bar{Z}_{n,j} \xrightarrow{\mathbb{P}} \eta_j$ for each j. Since d is fixed, the union bound and $\|v\|_2 \le \sqrt{d}\max_j |v_j|$ give

$$\mathbb{P}_P(\|\bar{Z}_n - \eta\|_2 > \varepsilon) \le \sum_{j=1}^{d} \mathbb{P}_P(|\bar{Z}_{n,j} - \eta_j| > \varepsilon/\sqrt{d}) \to 0.$$

Differentiability at η gives $h(\eta + u) - h(\eta) = a^{\mathrm{T}}u + r(u)$, where $r(u)/\|u\|_2 \to 0$ as $u \to 0$. Thus $r(\bar{Z}_n - \eta)/\|\bar{Z}_n - \eta\|_2 = o_P(1)$. The multivariate CLT gives $\sqrt{n}(\bar{Z}_n - \eta) \rightsquigarrow N_d(0, \Sigma)$, so $\sqrt{n}\|\bar{Z}_n - \eta\|_2 = O_P(1)$. Multiplying these two bounds shows that $\sqrt{n}\, r(\bar{Z}_n - \eta) = o_P(1)$. Consequently,

$$\sqrt{n}\{h(\bar{Z}_n) - h(\eta)\} = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} a^{\mathrm{T}}(Z_i - \eta) + o_P(1) \rightsquigarrow N(0, a^{\mathrm{T}}\Sigma a),$$

by the scalar CLT and Slutsky's theorem.

Covariance convergence in operator norm. Finite second moments imply

$$\mathbb{E}_P|Z_j Z_k| \le (\mathbb{E}_P Z_j^2\, \mathbb{E}_P Z_k^2)^{1/2} < \infty.$$

The LLN therefore applies to every entry of $Z_i Z_i^{\mathrm{T}}$. Using $\widehat{\Sigma}_n = n^{-1}\sum_i Z_i Z_i^{\mathrm{T}} - \bar{Z}_n \bar{Z}_n^{\mathrm{T}}$ and the convergence of the sample mean, we obtain $(\widehat{\Sigma}_n - \Sigma)_{jk} \xrightarrow{\mathbb{P}} 0$ for each j, k. Let $D_n = \widehat{\Sigma}_n - \Sigma$. For every $\varepsilon > 0$,

$$\mathbb{P}_P\left(\max_{j,k} |(D_n)_{jk}| > \varepsilon\right) \le \sum_{j,k=1}^{d} \mathbb{P}_P(|(D_n)_{jk}| > \varepsilon) \to 0.$$

The sum has only the fixed number d^2 of terms. Since $\|D_n\|_{\mathrm{op}} \le d\max_{j,k} |(D_n)_{jk}|$, we conclude that $\|\widehat{\Sigma}_n - \Sigma\|_{\mathrm{op}} \xrightarrow{\mathbb{P}} 0$. No fourth-moment assumption is needed.

Consistency of the estimated variance. Continuity of the gradient and Euclidean convergence of $\bar{Z}_n$ give $\|\widehat{a}_n - a\|_2 \xrightarrow{\mathbb{P}} 0$, where $\widehat{a}_n = \nabla h(\bar{Z}_n)$. In particular, $\|\widehat{a}_n\|_2 = O_P(1)$. The operator-norm inequality $|u^{\mathrm{T}} A v| \le \|u\|_2 \|A\|_{\mathrm{op}} \|v\|_2$ yields

$$|\widehat{\sigma}_n^2 - \sigma^2| \le \|\widehat{a}_n\|_2^2 \|\widehat{\Sigma}_n - \Sigma\|_{\mathrm{op}} + \|\Sigma\|_{\mathrm{op}} \|\widehat{a}_n - a\|_2 (\|\widehat{a}_n\|_2 + \|a\|_2) \xrightarrow{\mathbb{P}} 0.$$

This proves the scalar variance-consistency claim. □

B.2 Bootstrap for smooth functions of means

Proof of Theorem 3.9. Apply the argument of Theorem 3.5 to every fixed linear combination of Z_i. Conditional Cramér–Wold gives the multivariate bootstrap CLT. Conditionally, $\bar{Z}_n^* - \bar{Z}_n = O_{P^*}(n^{-1/2})$ in P-probability. Taylor expansion about $\bar{Z}_n$ gives

$$\sqrt{n}\{h(\bar{Z}_n^*) - h(\bar{Z}_n)\} = \nabla h(\bar{Z}_n)^{\mathrm{T}} \sqrt{n}(\bar{Z}_n^* - \bar{Z}_n) + o_{P^*}(1).$$

Continuity of the gradient controls this remainder in a fixed neighborhood of η. The probability of leaving that neighborhood vanishes. Since $\nabla h(\bar{Z}_n) \xrightarrow{\mathbb{P}} a$, the conditional limit has variance $a^{\mathrm{T}} \Sigma a$. Applying Lemma D.1 to each coordinate Z_j and product $Z_j Z_k$ gives consistency of the recomputed covariance and hence of the bootstrap scale. Conditional Slutsky (Lemma D.2) and Lemma 3.7 complete the proof. □

C Proof of the subsampling theorem

C.1 Proof of the main result

First center each subsample statistic at the unknown θ and define

$$U_{n,b}(t) = \binom{n}{b}^{-1} \sum_{|I|=b} \mathbf{1}\{\tau_b(\widehat{\theta}_{b,I} - \theta) \le t\}.$$

The observed centering is then a small common shift of its argument.

Proof of Theorem 5.1. For fixed t, iid sampling implies $\mathbb{E}_P U_{n,b}(t) = \mathbb{P}_P\{\tau_b(\widehat{\theta}_b - \theta) \le t\} \to J(t)$ because $b \to \infty$ and J is continuous. Lemma C.1 gives $\mathrm{Var}_P\{U_{n,b}(t)\} \le [4\lfloor n/b \rfloor]^{-1} \to 0$. Thus $U_{n,b}(t) \xrightarrow{\mathbb{P}} J(t)$. Lemma 3.7 gives uniform convergence.

Let $\delta_n = \tau_b(\widehat{\theta}_n - \theta)$. Since the full-sample root converges in distribution, it is tight, and $\tau_b/\tau_n \to 0$ implies $\delta_n = o_P(1)$. Exactly,

$$L_{n,b}(t) = U_{n,b}(t + \delta_n).$$

A continuous cdf on $\mathbb{R}$ is uniformly continuous. Consequently,

$$\|L_{n,b} - J\|_\infty \le \|U_{n,b} - J\|_\infty + \sup_t |J(t + \delta_n) - J(t)| \xrightarrow{\mathbb{P}} 0.$$

Quantile inversion uses the argument of Proposition 4.1, with τ_n in place of $\sqrt{n}$. □

C.2 Supporting lemma: overlap is controlled by partitions

Lemma C.1 (Variance of a bounded subset average). *For iid observations, a symmetric kernel h_b taking values in $[0,1]$, and $U = \binom{n}{b}^{-1} \sum_{|I|=b} h_b(X_I)$, let $k = \lfloor n/b \rfloor$. Then $\mathrm{Var}_P(U) \le 1/(4k)$.*

Proof of Lemma C.1. Independently permute the n indices uniformly and divide the first kb into k disjoint blocks of size b. Let V be the average of the k block-kernel values. Every block is a uniform size-b subset, so $\mathbb{E}(V \mid \mathcal{D}_n) = U$. Unconditionally, the block-kernel values are independent and identically distributed: disjoint groups of iid observations remain independent under a random permutation independent of their values. Each has variance at most $1/4$, so $\mathrm{Var}(V) \le 1/(4k)$. The law of total variance gives $\mathrm{Var}(U) \le \mathrm{Var}(V)$. □

This proof does not assume that all overlapping subsets are independent. A naive count of all overlaps would demand unnecessarily strong restrictions on b.

D Auxiliary lemmas

D.1 Proof of the cdf convergence lemma

Proof of Lemma 3.7. For any $\varepsilon > 0$, choose finite endpoints with H-tail mass below ε, then a finite ordered grid between them on which each increment of H is below ε. Monotonicity traps $H_n(t)$ between adjacent grid values. The supremum error is at most the largest grid error plus ε, with analogous endpoint bounds in the tails. Finitely many pointwise convergences control the grid. For random cdfs use a union bound over the same finite grid. □

Lemma D.1 (Empirical-bootstrap LLN under integrability). *Let $Y_i = g(X_i)$ be iid with $\mathbb{E}_P|Y| < \infty$. Under empirical resampling, $P_n^* g - P_n g = o_{P^*}(1)$ in P-probability, and hence $P_n^* g \to Pg$ in the same conditional sense.*

Proof of Lemma D.1. Truncate Y to $Y^{(K)} = \max(-K, \min(Y, K))$. For fixed K, the conditional variance of the bootstrap mean of $Y^{(K)}$ is at most K^2/n, so conditional Chebyshev gives concentration about its empirical mean. For the tail, conditional Markov bounds the probability of a bootstrap absolute tail average exceeding ε by $P_n|Y - Y^{(K)}|/\varepsilon$. Its expectation under the original sample is $\mathbb{E}|Y - Y^{(K)}|/\varepsilon \to 0$ as $K \to \infty$. The original empirical tail is controlled in the same way. First let $n \to \infty$ with fixed K, then $K \to \infty$. Finally $P_n g \xrightarrow{\mathbb{P}} Pg$ by the ordinary LLN. □

Lemma D.2 (Conditional Slutsky). *Let $U_n^* \in \mathbb{R}^d$ and $V_n^* \in \mathbb{R}^k$ satisfy*

$$\mathcal{L}(U_n^* \mid \mathcal{D}_n) \rightsquigarrow \mathcal{L}(U) \quad \text{in } P\text{-probability}, \qquad \mathbb{P}(\|V_n^* - c\|_2 > \varepsilon \mid \mathcal{D}_n) \xrightarrow{\mathbb{P}} 0$$

for every $\varepsilon > 0$, where $c \in \mathbb{R}^k$ and the law of U are deterministic. Then the conditional law of (U_n^*, V_n^*) converges weakly to that of (U, c) in P-probability. Consequently, for any map g continuous at (U, c) almost surely,

$$\mathcal{L}\{g(U_n^*, V_n^*) \mid \mathcal{D}_n\} \rightsquigarrow \mathcal{L}\{g(U, c)\} \quad \text{in } P\text{-probability}.$$

In particular, for scalar variables and $c \neq 0$, U_n^*/V_n^* has conditional weak limit U/c; define the ratio arbitrarily when $V_n^* = 0$. No conditional independence between U_n^* and V_n^* is required.

Proof of Lemma D.2. For a bounded-Lipschitz function f in the unit class and any $\delta > 0$,

$$|\mathbb{E}\{f(U_n^*, V_n^*) - f(U_n^*, c) \mid \mathcal{D}_n\}| \le \delta + 2\mathbb{P}(\|V_n^* - c\|_2 > \delta \mid \mathcal{D}_n).$$

This bound holds uniformly over f. Also, $u \mapsto f(u, c)$ is in the bounded-Lipschitz unit class, so the assumed conditional weak convergence controls the difference between $\mathbb{E}\{f(U_n^*, c) \mid \mathcal{D}_n\}$ and $\mathbb{E}f(U, c)$ uniformly over f. First let $n \to \infty$, then $\delta \downarrow 0$, to obtain joint conditional weak convergence. Applying the continuous mapping theorem along almost-sure subsequences, as in Appendix A, proves the claim for g. When $c \neq 0$, the conditional probability that $V_n^* = 0$ is bounded by $\mathbb{P}(|V_n^* - c| > |c|/2 \mid \mathcal{D}_n) \xrightarrow{\mathbb{P}} 0$. □

E Additional numerical details

The figures use exact distribution formulas and fixed parameters, so no random seed or simulation error is involved. For exponential mean one, $n\bar{X} \sim \mathrm{Gamma}(n, 1)$, with shape n and scale one. Thus, for $x \ge -\sqrt{n}$,

$$\mathbb{P}\{\sqrt{n}(\bar{X} - 1) \le x\} = F_{\mathrm{Gamma}(n,1)}(n + x\sqrt{n}),$$

and the cdf is zero below that support. Figure 2 takes $n = 25$ and compares this cdf with $\Phi(x)$ and $\Phi(x) + (1 - x^2)\phi(x)/(3\sqrt{n})$.

For $\theta = 1$, the exact endpoint-root cdf is zero for $t < -n$, equals $(1 + t/n)^n$ on $[-n, 0]$, and equals one for $t > 0$. Figure 1 takes $n = 40$. The arrow at zero indicates the limiting bootstrap atom $1 - e^{-1} \approx 0.6321$; it is not an estimate from data.

The assignment example enumerates all six combinations. The unequal-variance exercise gives $V_{\mathrm{actual}} = 9$ and $V_{\mathrm{perm}} = 13.5$. The studentized Cornish–Fisher correction gives 1.21745 to five decimal places.

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